**Problem Set for 13 March 2025**

**Exercise 1** Let $f : U \to V$ be linear map and $\dim U = n$, $\dim V = m$.

1. Let $E \subset V$ be an $k$-dimensional subspace, prove that
$$\dim f^{-1}(E) \geq n - m + k.$$
2. When $f$ is surjective, the equality holds.

**Exercise 2** Read the following table and briefly talk about your understanding or confusion?

| **Concepts** | **Matrices** | **Linear maps** |
|---|---|---|
| Linear transformation | $A_{m\times n}$ | $\varphi : \mathbb{F}^m \to \mathbb{F}^n$ |
| Basis | $(u_1, \ldots, u_m)$, an invertible matrix | Omit. |
| Image of a basis | $(u_1, \ldots, u_m) \cdot A$ | $(\varphi(u_1), \ldots, \varphi(u_m))$ |
| | $E_{i,j}$ | $\varphi(u_i) = v_j$, and $\varphi(u_{\neq i}) = 0$ |
| Sum of maps | $A_1 + A_2$ | $\varphi_1 + \varphi_2$ |
| Scalar of maps | $\lambda \cdot A$ | $\lambda \cdot \varphi$ |
| Composition | $A \cdot B$ | $\psi \circ \varphi$ |
| V-composition | $\binom{A}{B}$ | $\begin{array}{ccc} U & \to & V \times W \\ u & \mapsto & (\varphi(u), \psi(u)) \end{array}$ |
| H-composition | $(A\ B)$ | $\begin{array}{ccc} U \times V & \to & W \\ (u, v) & \mapsto & \varphi(u) + \psi(v) \end{array}$ |
| D-composition | $\begin{pmatrix} A & O \\ O & B \end{pmatrix}$ | $\begin{array}{ccc} U_1 \times U_2 & \to & V_1 \times V_2 \\ (u_1, u_2) & \mapsto & (\varphi(u_1), \psi(u_2)) \end{array}$ |
| Dual map | $A^T$ | $\begin{array}{ccc} \mathrm{Hom}_{\mathbb{F}}(V, \mathbb{F}) & \to & \mathrm{Hom}_{\mathbb{F}}(U, \mathbb{F}) \\ f & \mapsto & f \circ \varphi \end{array}$ |
| Inverse | $A^{-1}$ | $\varphi^{-1}$ |

| **Concepts** | **Matrices** | **Linear maps** |
|---|---|---|
| Adj | $\mathrm{Adj}(A)$ | $\begin{array}{ccc} \bigwedge^{n-1} U & \to & \bigwedge^{n-1} U \\ \sum u^\lambda_{i_1} \wedge \cdots \wedge u^\lambda_{i_{n-1}} & \mapsto & \sum \varphi(u^\lambda_{i_1}) \wedge \cdots \wedge \varphi(u^\lambda_{i_{n-1}}) \end{array}$ |
| Rank | $r(A)$ | $\dim \operatorname{im} \varphi$ |
| Equivalence | $A = P^{-1}\widetilde{I}_r Q$ | $\exists \, \{u_i\}_{i=1}^m, \{v_j\}_{j=1}^n : \varphi(u_{i\le r}) = v_i$, otherwise $0$. |
| Injection | Full row rank | Omit. |
| | $\exists B, AB = I$ | Omit |
| Surjection | Full column rank | Omit. |
| | $\exists B, BA = I$ | Omit. |
| Fundamental decomposition | $A = R \cdot C$, where $R$ has full row rank, $C$ has full column rank | $\varphi$ is always the composition of a surjection (from its domain) and an injection (to its codomain). |
| Elementary transformation | $S_{i,j}$ | $\varphi$ swaps $u_i$ and $u_j$ |
| | $I + \lambda E_{i,j}$ | $\varphi(u_j) = v_j$ for $j \neq i$, and $\varphi(u_i) = u_i + \lambda u_j$. |
| | $I + (\lambda - 1)E_{i,i}$ | $\varphi(u_j) = v_j$ for $j \neq i$, and $\varphi(u_i) = \lambda v_i$. |
| Projection | Take the first $r$ columns | $\pi \circ \varphi$, where $\pi$ is the projection $\mathrm{span}(\{v_i\}_{i=1}^n) \to \mathrm{span}(\{v_i\}_{i=1}^r)$. |
| Inclusion | Add $k$ zero columns | $\iota \circ \varphi$, where $\iota$ is the inclusion $\mathrm{span}(\{v_i\}_{i=1}^n) \to \mathrm{span}(\{v_i\}_{i=1}^{n+k})$. |
| Nilpotent | $A^N = O$ | Omit |
| Similarity | $A = P^{-1}BP$ | $\exists$ base change $(u_i) = (v_i) \cdot P$, such that the linear map has matrix form $B$. |
| Invariant subspace | $A \sim \begin{pmatrix} A_1^{(r\times r)} & A_2 \\ O & A_4 \end{pmatrix}$ | $\exists U' \subset U : \varphi(U') \subset U'$. |
| Quotient an invariant subspace | $\begin{pmatrix} A_1^{(r\times r)} & A_2 \\ O & A_4 \end{pmatrix} \mapsto A_4$ | $\varphi(x + U) = \varphi(x) + U$. |

| Concepts | Matrices | Linear maps |
| --- | --- | --- |
| Mutual annihilation | $AB = O_{m\times m}$, and $BA = O_{n\times n}$ | See example class last year. |
| Blocks | $A \sim \begin{pmatrix} A_1 & O \\ O & A_2 \end{pmatrix}$ | $U_1 \oplus U_2 = U$, such that $\varphi(U_i) \subset U_i$. |
| Upper triangulation | $A \sim T$ | There exists a flag of $U_i$ with $U_i \subset U_{i+1}$, $\dim U_i = i$, such that $\varphi(U_i) \subset U_i$. |
| Diagonalisation | $A \sim P^{-1}\Lambda P$ | Omit. |
| Fitting lemma | $A \sim \begin{pmatrix} N & O \\ O & M \end{pmatrix}$, where $N$ is nilpotent, and $M$ is invertible. | For $\varphi : U \to U$, $\mathrm{im}(\varphi^N)$ and $\ker(\varphi^N)$ are invariant subspaces for large $N$. In particular, $U = \mathrm{im}(\varphi^N) \oplus \ker(\varphi^N)$. |
| Tensor | Kronecker product | Tensor product |

**Exercise 3 (optional)** Let $f, g : \mathbb{C}^{2024} \to \mathbb{C}^{2025}$ be linear injections, such that $\mathrm{im}(f) + \mathrm{im}(g) = \mathbb{C}^{2025}$.

1. One can find $u, v$ such that $f(u) \notin \mathrm{im}(g)$ and $g(v) \notin \mathrm{im}(f)$. Show that $\{u, v\}$ is a linearly independent set.
2. Find bases such that the matrix forms of $f$ and $g$ are

$$f \mapsto \begin{pmatrix} I_{2024} \\ \mathbf{0}^T \end{pmatrix}, \quad g \mapsto \begin{pmatrix} \mathbf{0}^T \\ I_{2024} \end{pmatrix}.$$

3. Say $(\mathbb{C}^{2024} \xrightarrow{\alpha} \mathbb{C}^{2024}; \mathbb{C}^{2024} \xrightarrow{\beta} \mathbb{C}^{2025})$ is a good pair of linear endomorphisms, provided

$$\beta \circ (af + bg) = (af + bg) \circ \alpha, \quad (\forall a, b \in \mathbb{C}).$$

   Prove that $(\alpha; \beta)$ forms a linear space, and find its dimension.